

represent different letters. The scale cannot be changed. Braille dots are spaced 2.5mm apart within a cell, and cell-to-cell pitch is about 6.25mm.

D'Souza needed to create a mechanical system that would drive the pins according to the data and, for this, he used actuators. Electronic signals drive the actuators, each of which drives a pin on Braille display. Sounds simple, but try to implement it and the complications begin.

First and foremost, the display has to show enough data for a person to read. Touche has a 20-character display, with each character comprising dots or pins arranged in three rows of two columns each; this means fitting in 120 actuators within one line. When the reader feels the data, pins have to withstand a pressure of 15 to 20 grams.

Small solenoids and motors cannot bear this pressure, others are too big, consume huge current, or are too costly. D'Souza came up with the idea of using appropriately-positioned micro-motors, the ones used as vibrators in mobile phones that cost around ₹ 15 each.

But with these being about 4mm in diameter, size and stability became serious issues. The trick was to use these only to indicate logical states and distribute and stack these in other available free space. To shift the selection from the motors to the pins on the display, he used a system of connecting levers. To refresh data and load in the next set, he designed a system of mass lifting.

How Touche was born

With the mechanical system used, converse operation turned out to be easy. D'Souza added a Braille keyboard to make sure the visually-challenged could also write and not just read digital data.

Touche is a refreshable Braille display. All it needed was a smart combination of electronic and mechanical principles, many of which

Igniting the Spark of Inspiration



"I was listening to an orchestra play the music of a hymn from the movie Titanic as the ship began to sink. While listening to the music, I was running my finger over the music score and trying to co-relate what I was hearing with the dots and lines on the score. While doing that, I felt inspired with the thought that if the dots could come out of the page, someone who is blind would be able to read music."

"The same day, I listened to someone speak about the exorbitant cost of printing Braille literature and the huge amount of storage space required for Braille matter when compared to storage of normal books. This led me to try and figure out how I could make a device that would produce Braille dots using digital information. I did think of a few solutions but my ideas were dismissed as being impractical when reduced to Braille-size specifications. I used their discouragement as a challenge and produced a display prototype to prove that it could be done."

—Paul Gerard D'Souza is a freelancer with 20 years of experience in inventing. His inventions range from agro-machines to watch escapements, perpetual calendar, large date mechanisms and other inventions for the visually-impaired



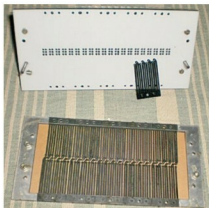
Side view of the motor assembly under the pin plate

the inventor was right to patent.

The device can now take in digital data, store it in its memory, access it line-by-line and convert it to Braille. Once refreshed with a simple press of a button, the next set of 20 characters is loaded. With all procedures involved for a line display, refreshing takes less than 100 milliseconds.

Touche costs one-tenth of other similar devices. It has been built using easily-available and serviceable components. Most importantly, all key components have been manufactured using locally-available workshop facilities.

Ask D'Souza and he says that the biggest challenge lay in convincing fabricators and producers of components, of the need to stick to specifications and the levels of accuracy he desired. With his unconventional approach, willingness to invest in something new with an eye for detail, overcoming mind-blocks and getting people's support turned



Pin-plate assembly: single module and complete assembly

out to be difficult. It took him six years to finally achieve his vision.

Touche has been demonstrated at Centre for Development of Advanced Computing (C-DAC), and Kanthari, a branch of Braille Without Borders, in Trivandrum, Kerala. It has found appreciation from visually-challenged users and was displayed to the public at a Mini Maker Faire in Bengaluru.

The designer aims to convert the set-up to use a single-board computer, to make it compatible with all software and input formats. With a few other enhancements in mind, he intends to release the product in the market in early 2016 and sincerely hopes to make the lives of the visually-impaired a lot easier and better. **EFY**